

Week 1

Activity Content

Description

Introduction to Statistics

In an attempt to study the problem of failure in field-installed computer equipment, data is collected on fifty field trips made to pair equipment. The random variables studied are X and the time in hours required to locate and rectify the problem.

These data are obtained:

1.52, 1.83, 2.25, 4.73, 2.89,
1.49, 1.34, 2.15, 2.66, 2.79, 1.35, 1.54, 4.59, 4.27, 3.91, 2.76, 3.03, 3.52, 5.97, 1.45,
3.07, 2.18, 1.38, 2.04, 1.49, 1.11, 1.24, 4.84, 2.82, 3.16, 4.58, 3.28, 1.30, 3.01, 1.20, 3.42, 1.86, 3.49, 3.9
3, 2.56, 2.63, 5.60, 4.60, 5.34, 1.62, 2.82, 4.88, 2.04, 1.62, .24.

- a. Construct a relative frequency histogram for the data (use six categories). Is X approximately normally distributed?
- b. Find the sample mean, variance, and standard deviation of X.
- c. Construct cumulative frequency Ogive for X, and use this Ogive to estimate the median of the data.
- d. Draw the box plot and stem and leaf plot for the data.

Tips for writing posts:

- Read the prompt and instructions carefully.
- Identify the most important words or phrases in the prompt to help you stay on point.
- Ensure your post is free from plagiarism. When alluding to an idea or text/image that is not your own, add a citation that identifies the full name of the source, the date it was published, and any other citation element.
- Use a formal writing style.
- Double-check your post to ensure there are no spelling mistakes and grammatical issues.

Tips for responding to peer responses:

Responding to other participants is optional and does not affect your grades. However, responding to your peers' posts is recommended and encouraged.

While responding to your peers' posts

- Provide constructive comments.
- Provide reasons for your agreement or disagreement.

For constructing your arguments, we recommend that you use steelmanning, which implies that you find the best argument to test an opposing opinion. We do not recommend that you use strawman arguments. A strawman is a fallacious argument that distorts an opposing stance to make it easier to attack.

Click for more information about [Strawman Arguments](#).

Click for more information on [Knocking Down a Steel Man: How to Argue Better](#).